

Khawaja Yunus Ali University

School of Science and Engineering
Department of Computer Science and Engineering

Final Examination: Spring – 2021

Program: B.Sc. in CSE (Batch – 11th) 1st Year 2nd Semester

Course Title: Physics-II

Course Code: PHY 1201

Time: 10 Minutes

Total Marks: 10 X 1 = 10

Student ID:	Batch:	Session:
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Part 1: MCQ

These are single best answer type of questions. Each of the following question or statement has five suggested answers or completions. Write down the most appropriate one on your answer script.

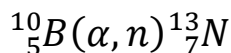
- What is the angle between electric and magnetic field in light?
 - 0°
 - 30°
 - 45°
 - 60°
 - 90°
- Which is a unitless quantity?
 - Fringe width
 - Wavelength
 - Amplitude
 - Refractive index
 - Grating Constant
- The phase difference between two coherent sources should be _____.
 - zero
 - π
 - 2π
 - $\pi/2$
 - $3\pi/4$
- The bending of light into the geometrical shadow is called _____.
 - reflection
 - refraction
 - interference
 - diffraction
 - polarization
- The number of lines on the grating is 14000 per inch, what will be the grating constant?
 - $0.71 \times 10^{-4} \text{cm}$
 - $1.81 \times 10^{-4} \text{cm}$
 - $2.54 \times 10^{-4} \text{cm}$
 - $4.62 \times 10^{-4} \text{cm}$
 - $7.11 \times 10^{-4} \text{cm}$

6. To happen the photoelectric effect, what condition be essential between incident photon energy (E) and work function (W_0)?
- a) $E < W_0$
 - b) $E \ll W_0$
 - c) $E \leq W_0$
 - d) $E \gg W_0$
 - e) $E \geq W_0$
7. Charge of the alpha particle is _____ .
- a) 0.53 C
 - b) 1.6×10^{-19} C
 - c) 3.2×10^{-19} C
 - d) 1.67×10^{-27} C
 - e) 9.1×10^{-31} C
8. Energy released per fission reaction of ^{235}U is _____ .
- a) 13.6 eV
 - b) 17.6 eV
 - c) 200 MeV
 - d) 931 MeV
 - e) 1172 MeV
9. What is the value of the Compton wavelength?
- a) $0.53 \text{ \AA}$
 - b) $0.053 \text{ \AA}$
 - c) $2.4 \text{ \AA}$
 - d) $0.24 \text{ \AA}$
 - e) $0.024 \text{ \AA}$
10. In the nuclear reaction, proton is denoted by _____ .
- a) ${}^4_2\text{He}$
 - b) 1_0n
 - c) 1_0p
 - d) ${}^1_1\text{H}$
 - e) ${}^2_1\text{H}$

(Answer Question No. 1 and any four from the rest.)

[The figures in the right margin indicate full marks for the respective question]

1. See the stem and answer the questions that follows:

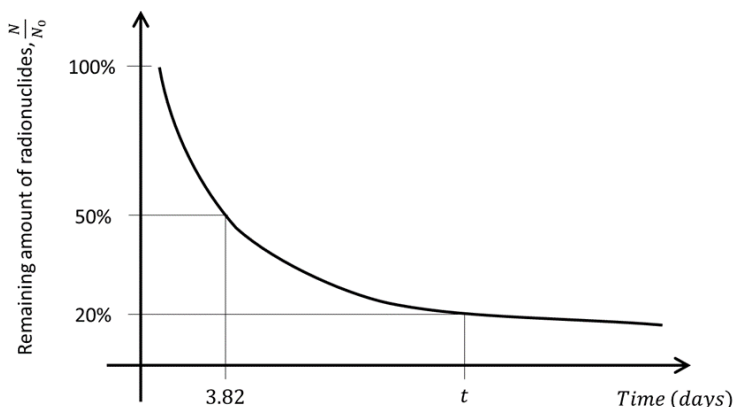


Short form of a nuclear reaction

(Where, Mass of proton = 1.00728 a.m.u., Mass of neutron = 1.00867 a.m.u., Mass of the helium nucleus = 4.00276 a.m.u. and 1 a.m.u. = 931.5 MeV)

- a) Write the complete form of the reaction. 1
 - b) Calculate the radius of the product nucleus. 2
 - c) Compare the binding energies of the incident and outgoing particles. 3
- a) Justify the statement that “The force is invariant in Galilean transformation”. 3
 - b) A spacecraft is moving relative to the earth. An observer on the earth finds that, according to her clock, 3601sec elapse between 1 p.m. and 2 p.m., on the spacecraft’s clock. What is the spacecrafts speed relative to the earth? 3

3. See the radioactive decay curve for a substance and answer the questions that follows:



- a) What is the half-life of radioactive substance? 1
 - b) Evaluate the value of mean life of above mentioned substance. 2
 - c) Find the value of t from the graph. 3
- a) State Huygens Principle. 1
 - b) How Newtons rings are formed? 2
 - c) Light from a sodium vapour lamp ($\lambda = 589 \text{ nm}$) forms an interference pattern on a screen 0.8 m from a pair of slits. The bright fringes in the pattern are 0.35 cm apart. What is the slit separation? 3
- a) Explain the Fraunhofer class of diffraction. 2
 - b) What is the resolving power of grating? Calculate the minimum number of lines in a grating which just resolve the sodium lines in the first order spectrum. The wavelengths are 5890 A.U. and 5896 A.U. 1+3=4
- a) Write a short note on “Nicol Prism”. 2
 - b) How will you orient the polarizer and the analyzer so that a beam of natural light is reduced to – (i) 0.125 (ii) 0.25 (iii) 0.5 and (iv) 0.75 of its original intensity? 4